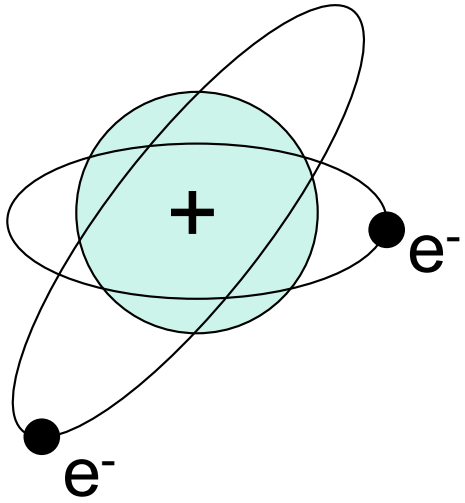
Lectures 8

Conductors

Dr. Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard, Brian Cunningham, and Victoria Shao
- Copying, publishing or distributing without express written permission is prohibited.

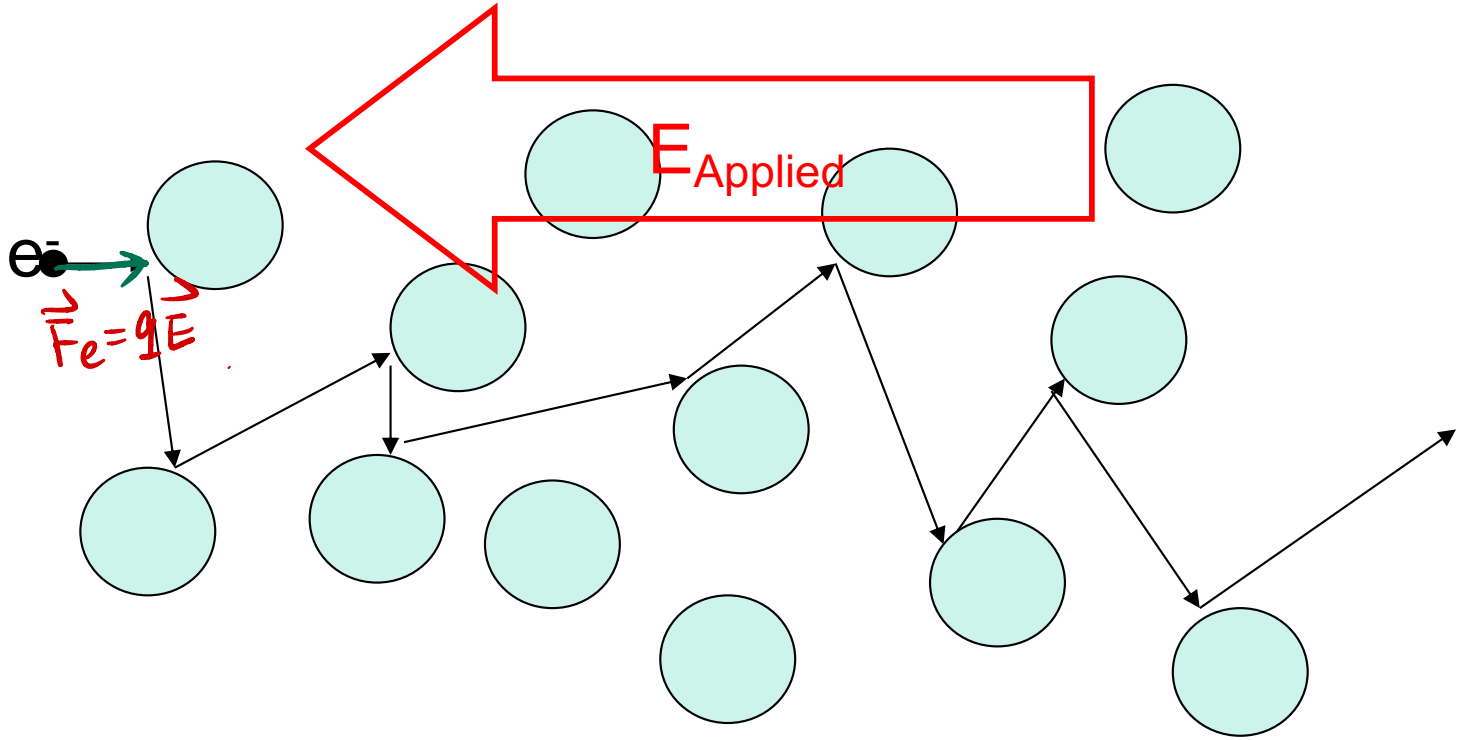
Atomic Model of Conductivity



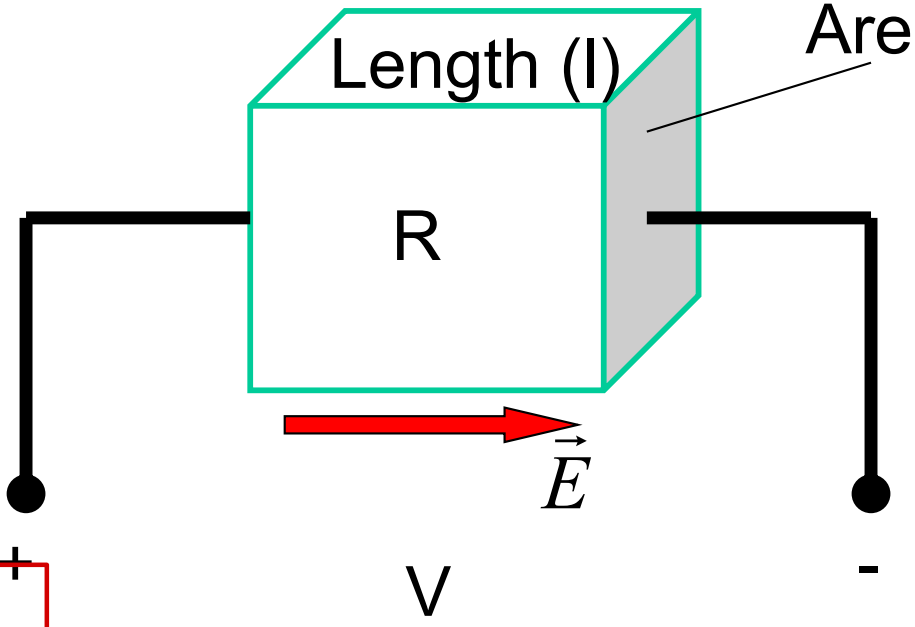
- Tightly bound inner orbitals
- Loosely bound outer orbitals
    - Free to escape the nucleus and move around inside the material

Conduction of Free Electrons

- The electron cannot travel in a straight path for very long
- It keeps running into the nuclei, getting deflected MANY times, meeting RESISTANCE



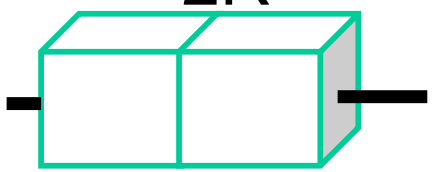
Resistors



$$R = \frac{V}{I}$$

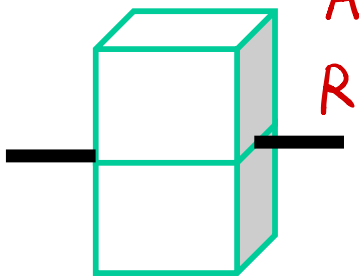
$$\Omega(Ohms) = \frac{volts}{amps}$$

$l \rightarrow 2l$
$R \rightarrow 2R$.



$$\left(\frac{1}{R} + \frac{1}{R}\right)^{-1}$$

$A \rightarrow 2A$
$R \rightarrow \frac{R}{2}$.



R tells us how much total resistance is encountered, but does not tell us anything fundamental about the resisting material

Resistivity

$$\rho = \text{Resistivity, units of } (\Omega \cdot \text{m})$$

- Tells us how much resistance the *material* provides, factoring out the dimensions of the resistor

$$R = \rho \frac{l}{A}$$

Conductivity

$$\sigma = \frac{1}{\text{Resistivity}} = \frac{l}{R \cdot A} = \frac{1}{\Omega \cdot m}$$

- More common to describe materials in terms of conductivity rather than resistivity
- High conductivity = low resistivity
- Special units:

$$\sigma = \frac{1}{\Omega \cdot m} = \frac{Siemens}{m} = \frac{S}{m}$$

Conductivity Table

	Material	Conductivity, σ m^{-1}
Insulators	Quartz, fused	$\sim 10^{-17}$
	Ceresin wax	$\sim 10^{-17}$
	Polystyrene	$\sim 10^{-16}$
	Sulfur	$\sim 10^{-15}$
	Mica	$\sim 10^{-15}$
	Paraffin	$\sim 10^{-15}$
	Rubber, hard	$\sim 10^{-15}$
	Porcelain	$\sim 10^{-14}$
	Glass	$\sim 10^{-12}$
	Bakelite	$\sim 10^{-9}$
	Distilled water	$\sim 10^{-4}$
insulators	Dry, sandy soil	$\sim 10^{-3}$
	Marshy soil	$\sim 10^{-2}$
	Fresh water	$\sim 10^{-2}$
	Animal fat‡	4×10^{-2}
Poor conductors	Animal muscle ($\perp$ to fiber)‡	0.08
	Animal, body (ave)‡	0.2
	Animal muscle ($\parallel$ to fiber)‡	0.35
	Animal blood	0.7
	Germanium (semiconductor)	~2
Conductors	Seawater	~4
	Ferrite	10^2
	Tellurium	$\sim 5 \times 10^2$

	Material	Conductivity, σ m^{-1}
Conductors	Silicon	10^3
	Carbon	$\sim 3 \times 10^4$
	Graphite	$\sim 10^5$
	Cast iron	$\sim 10^6$
	Mercury	10^6
	Nichrome	10^6
	Stainless steel	10^6
	Constantan	2×10^6
	Silicon steel	2×10^6
	German silver	3×10^6
	Lead	5×10^6
	Tin	9×10^6
	Phosphor bronze	10^7
	Brass	1×10^7
	Zinc	1.7×10^7
	Tungsten	1.8×10^7
	Duralumin	3×10^7
	Aluminum, hard-drawn	3.5×10^7
	Gold	4.1×10^7
	Copper	5.7×10^7
	Silver	6.1×10^7
Super-conductors	Hg (at <4.1 K)	∞
	Nb (at <9.2 K)	∞
	$Nb_3(Al\text{-}Ge)$ (at <21 K)	∞
	$YBa_2Cu_3O_7$ (at <80 K)	∞

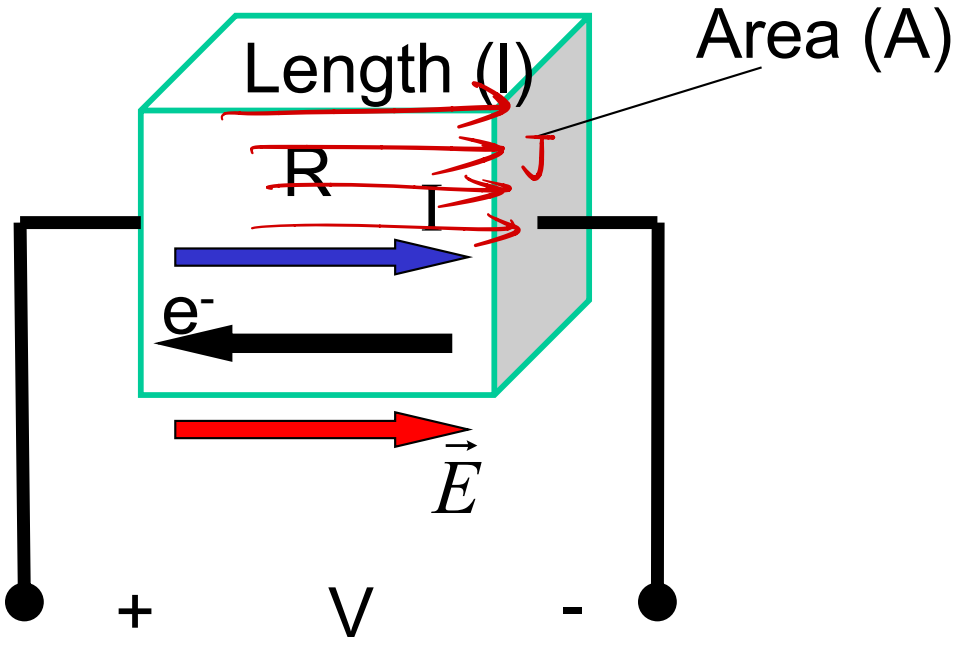
or low frequencies. At 20°C except where noted.

Ohm's Law

$$|E| = \frac{V}{l} \quad \left(\frac{V}{m}\right)$$

$$|J| = \frac{I}{A} \quad \left(\frac{A}{m^2}\right)$$

$$R = \frac{l}{\sigma A} \quad (\Omega)$$



$$V = IR$$

$$E \cdot l = (J \cdot A)\left(\frac{l}{\sigma A}\right) = \frac{1}{\sigma} J \cdot l$$

$$\boxed{\vec{J} = \sigma \vec{E}}$$

Ohm's Law in 329.

Maxwell's Equations in a Conductor

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

In free space

$$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

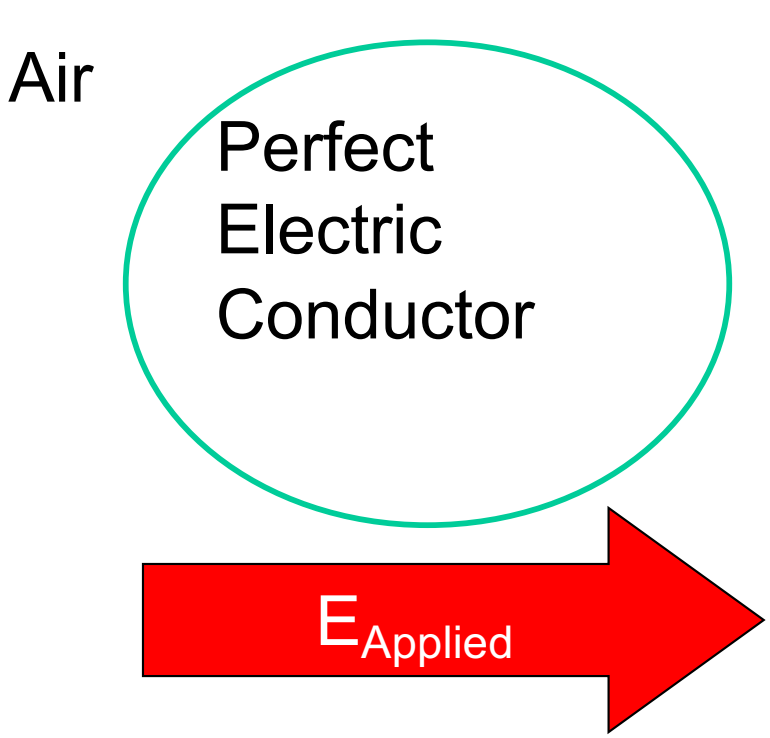
In a conducting material

Conduction current: Moving charge

Displacement Current: Time varying E-field

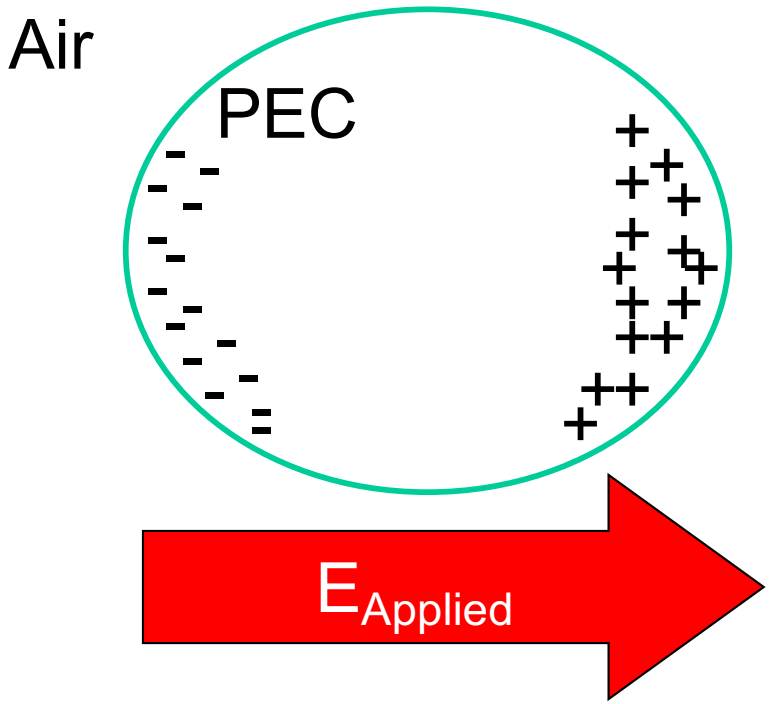
Perfect Electric Conductor (PEC) in an Applied Electric Field

$\sigma \to \infty$



What Happens?

PEC in an Applied Electric Field

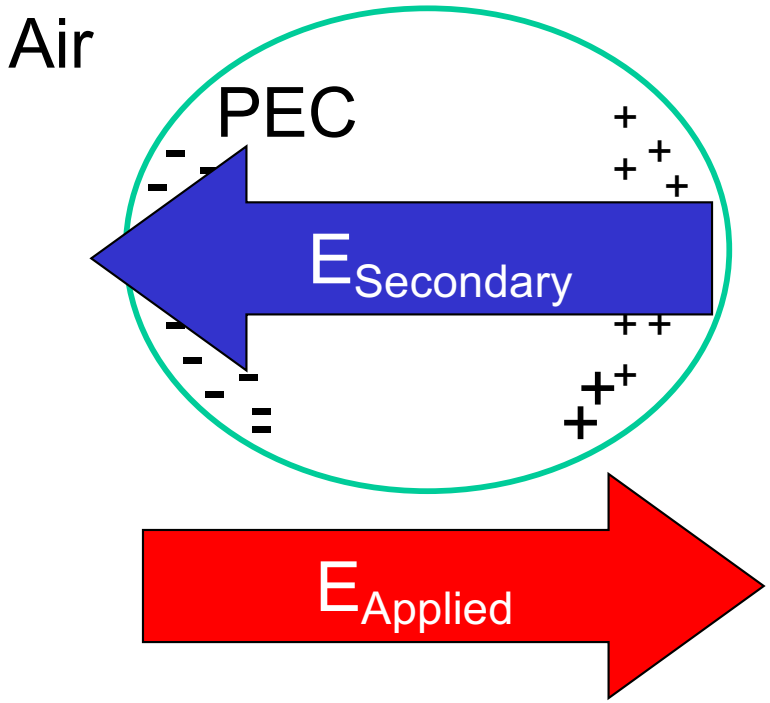


What Happens?

-Free e^- move in direction opposite of applied E

- When e^- moves away from its nucleus, it leaves a + charge behind
- The material as a whole is still NEUTRALLY CHARGED but the charge has now been redistributed

E=0 inside PEC



What Happens Next?

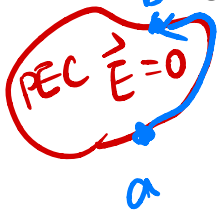
-The separated + and - charges create their own SECONDARY INTERNAL E field that *EXACTLY CANCELS* the applied E field

- The TOTAL E field inside the perfect conductor is ALWAYS ZERO

- If it weren't zero, free charge would continue drifting till it is!

Conductors and Dielectric

- $\sigma \to \infty$ corresponds to a perfect electrical conductor (PEC)
  - $\vec{\boldsymbol{E}} = \mathbf{0}$ inside independent of $\vec{\boldsymbol{J}}$
  - in analogy with $V = 0$ across a short circuit element

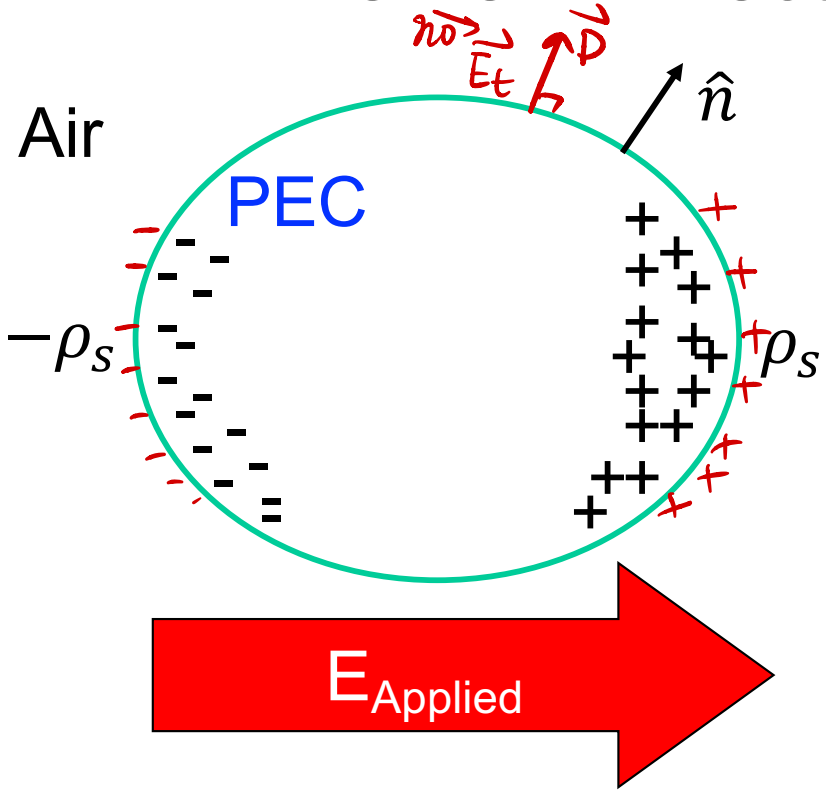


$V_{ab} = -\int_a^b \vec{E} \cdot d\vec{l} = 0.$

$\vec{\boldsymbol{E}} = \mathbf{0}$ in PEC unconditionally

- $\sigma \to 0$ corresponds to a perfect dielectric
  - $\vec{\boldsymbol{J}} = 0$ inside independent of $\vec{\boldsymbol{E}}$
  - in analogy with $I = 0$ through an open circuit element

Review: Electrostatic field and PEC



Given $\vec{E} = \mathbf{0}$

1. Potential V=constant on PEC surface.

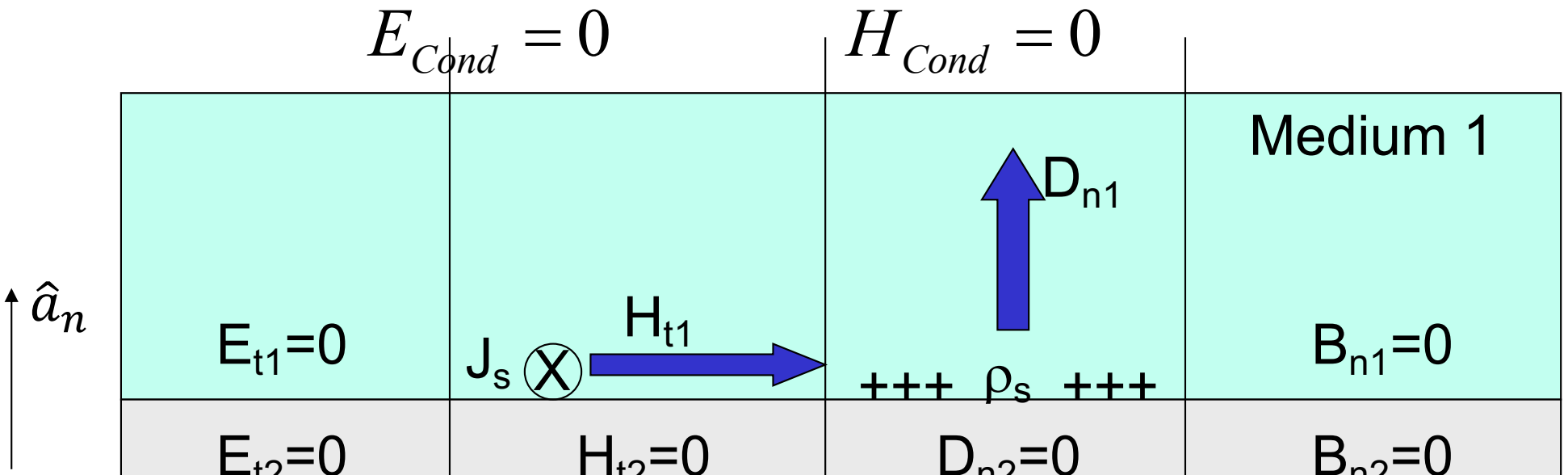
2. Gauss's Law

$$\rho = \nabla \cdot \vec{D} = \nabla \cdot \epsilon \vec{E} = 0$$

3. Boundary condition equations on PEC

$$\hat{n} \cdot \vec{D} = \rho_s \quad \text{and} \quad \hat{n} \times \vec{E} = 0$$

Surface of Perfect Conductor



$E_{t1} = 0$

$\hat{a}_n \times \vec{E}_1 = 0$

$|\vec{H}_{t1}| = |\vec{J}_s|$

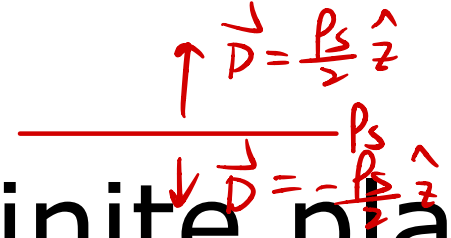
$\hat{a}_n \times \vec{H}_1 = \vec{J}_s$

$D_{n1} = \rho_s$

$\hat{a}_n \cdot \vec{D}_1 = \rho_s$

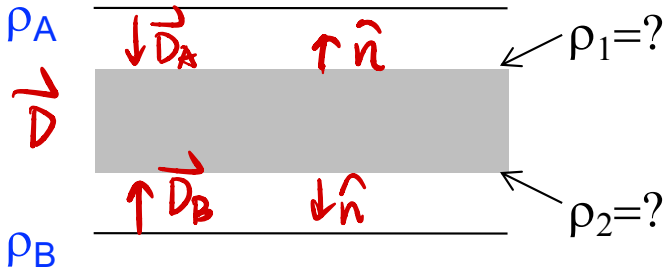
$B_{n1} = 0$

$\hat{a}_n \cdot \vec{B}_1 = 0$



Infinite plane conducting slab

- An electrically neutral infinite plane conducting slab lies between two infinite plane sheets of uniform surface charge density $\rho_A > 0$ and $\rho_B > 0$. Find the surface charge densities ρ_1 and ρ_2 on the two slab surfaces.



between ρ_A & ρ_B.

$$\vec{D} = \vec{D}_A + \vec{D}_B = -\frac{\rho_A}{2}\hat{z} + \frac{\rho_B}{2}\hat{z}$$

ρ_1 surface. $\hat{n}\cdot\vec{D} = \rho_1 = \hat{z}\cdot\left(\frac{\rho_B - \rho_A}{2}\hat{z}\right)$

$$= \frac{\rho_B - \rho_A}{2} \text{ C/m}^2.$$

ρ_2 surface $\hat{n}\cdot\vec{D} = \rho_2 = (-\hat{z})\cdot\left(\frac{\rho_B - \rho_A}{2}\hat{z}\right)$

$$= \frac{\rho_A - \rho_B}{2} \text{ C/m}^2.$$

If $\vec{E} = 0$ inside a PEC, how about $\vec{H}$?

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \Rightarrow \quad 0 = -\frac{\partial \vec{B}}{\partial t}$$

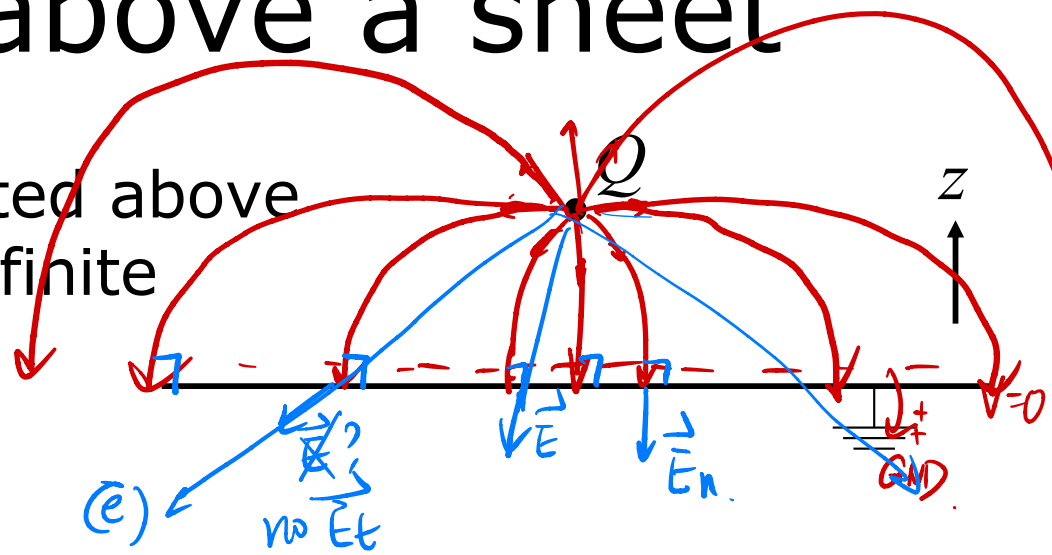
B and H must be STATIC

Technically, a non-zero static H-field can exist inside a PC, but how did it get there? It must have existed there for all time. But what created it in the first place? This is an ill-posed problem so in 329, we will assume $\vec{H} = \mathbf{0}$ inside a PEC.

$\vec{E} = \vec{D} = \vec{H} = \vec{B} = 0$

Challenge Question: Point charge above a sheet

- A positive charge is located above a perfectly conducting infinite ground plane at z=0



(a) **E**=0 above the plane

(b) **E**=0 below the plane shielding.

(c) A uniform charge density is induced on the plane $\hat{n}\cdot\vec{D}=\rho_s$

(d) The total induced charge on the plane is +Q $-Q$.

(e) The electric field lines have a radial component at the plane

$E_t=0 \rightarrow \hat{n}\times\vec{E}=0$. $\vec{E}\perp$ plane.

$\hat{n}\times(\vec{E}_1-\vec{E}_2)=0$

Lecture 8 Summary

- Ohm's Law: _____
- Conductivity: $\sigma = \frac{1}{\text{Resistivity}} = \frac{l}{R \cdot A}$
- Inside a perfect conductor, $\vec{\boldsymbol{E}} = ?$
- Next class
  - Dielectrics